Josephus Problem II

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2163 | Sorting and Searching | https://cses.fi/problemset/task/2163

Consider a game where there are n children (numbered 1,2,…,n) in a circle. During the game, repeatedly k children are skipped and one child is removed from the circle. In which order will the children be removed?

Input

The only input line has two integers n and k.

Output

Print n integers: the removal order.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $0 \le k \le 10^9$

Example

Input
7 2

Output
3 6 2 7 5 1 4